Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data to Delta Tables

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a Delta Live Table pipeline to load customer and order data from CSV files into Delta tables.

Target Cluster Configuration:

- Cluster Name: Databricks Cluster for DLT
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, spark-csv

Source Data Details:

- Dataset Name: Customer Data
- Location: `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`
- Format: CSV
- Description: Customer data in CSV format
- Dataset Name: Order Data
- Location: `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`
- Format: CSV
- Description: Order data in CSV format

Target Data Details:

- Output Name: Customer Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.customer`
- Format: Delta
- Description: Customer data in Delta format
- Output Name: Order Delta Table

- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.order`
- Format: Delta
- Description: Order data in Delta format

Job Flow / Pipeline Stages:

- Read customer CSV data and load into Delta table
- Read order CSV data and load into Delta table
- Add new column "TotalAmount" to order Delta table

Data Transformations / Business Logic:

- Step: Load customer data
- Description: Read customer CSV data and load into Delta table
- Transformation Logic: Use Spark CSV reader to read CSV data and write to Delta table
- Step: Load order data
- Description: Read order CSV data and load into Delta table
- Transformation Logic: Use Spark CSV reader to read CSV data and write to Delta table
- Step: Add TotalAmount column
- Description: Add new column "TotalAmount" to order Delta table
- Transformation Logic: Use Spark withColumn function to add new column as product of " PricePerUnit" and "Qty" columns

Error Handling and Logging:

- Log errors to a Delta table for auditing and debugging purposes
- Use try-except blocks to catch and handle exceptions during data loading and transformation

TR-DLT-002: Clean and Process Customer and Order Data

Related Functional Requirement(s): FR-DLT-002

Objective: Implement data quality checks to remove null and duplicate records from customer and order Delta tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.customer`
- Format: Delta
- Description: Customer data in Delta format
- Dataset Name: Order Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.order`

- Format: Delta
- Description: Order data in Delta format

Target Data Details:

- Output Name: Clean Customer Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.customer`
- Format: Delta
- Description: Clean customer data in Delta format
- Output Name: Clean Order Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.order`
- Format: Delta
- Description: Clean order data in Delta format

Job Flow / Pipeline Stages:

- Remove null records from customer and order Delta tables
- Remove duplicate records from customer and order Delta tables

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove null records from customer and order Delta tables
- Transformation Logic: Use Spark filter function to remove null records
- Step: Remove duplicate records
- Description: Remove duplicate records from customer and order Delta tables
- Transformation Logic: Use Spark dropDuplicates function to remove duplicate records

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-003: Create and Load ordersummary Table

Related Functional Requirement(s): FR-DLT-003

Objective: Implement a Delta Live Table pipeline to create and load ordersummary table with joined customer and order data using SCD Type 2.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Clean Customer Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.customer`
- Format: Delta
- Description: Clean customer data in Delta format
- Dataset Name: Clean Order Delta Table

- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.order`
- Format: Delta
- Description: Clean order data in Delta format

Target Data Details:

- Output Name: ordersummary Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary`
- Format: Delta
- Description: ordersummary data in Delta format

Job Flow / Pipeline Stages:

- Create ordersummary table if it does not exist
- Join customer and order Delta tables on CustId column
- Load joined data into ordersummary table using SCD Type 2

Data Transformations / Business Logic:

- Step: Join customer and order data
- Description: Join customer and order Delta tables on CustId column
- Transformation Logic: Use Spark join function to join customer and order data
- Step: Load data into ordersummary table
- Description: Load joined data into ordersummary table using SCD Type 2
- Transformation Logic: Use Spark SCD Type 2 implementation to load data into ordersummary table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-004: Update ordersummary Table on Customer Changes

Related Functional Requirement(s): FR-DLT-004

Objective: Implement a Delta Live Table pipeline to update ordersummary table whenever there are changes to customer data.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: Customer Delta Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.customer`
- Format: Delta
- Description: Customer data in Delta format

Target Data Details:

- Output Name: ordersummary Table

- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary`
- Format: Delta
- Description: ordersummary data in Delta format

Job Flow / Pipeline Stages:

- Identify changes to customer data
- Update ordersummary table by making old records inactive and new records active
- Update StartDate and EndDate columns accordingly to maintain history

Data Transformations / Business Logic:

- Step: Identify changes to customer data
- Description: Identify changes to customer data
- Transformation Logic: Use Spark join function to identify changes to customer data
- Step: Update ordersummary table
- Description: Update ordersummary table by making old records inactive and new records active
- Transformation Logic: Use Spark SCD Type 2 implementation to update ordersummary table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-005: Create and Load customeraggregatespend Table

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a Delta Live Table pipeline to create and load customeraggregatespend table with aggregated data from ordersummary table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary`
- Format: Delta
- Description: ordersummary data in Delta format

Target Data Details:

- Output Name: customeraggregatespend Table
- Location: `gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend`
- Format: Delta
- Description: customeraggregatespend data in Delta format

Job Flow / Pipeline Stages:

- Create customeraggregatespend table if it does not exist
- Aggregate TotalAmount column from ordersummary table by Name and Date columns

- Load aggregated data into customeraggregatespend table

Data Transformations / Business Logic:

- Step: Aggregate data
- Description: Aggregate TotalAmount column from ordersummary table by Name and Date columns
- Transformation Logic: Use Spark aggregate function to aggregate data
- Step: Load data into customeraggregatespend table
- Description: Load aggregated data into customeraggregatespend table
- Transformation Logic: Use Spark write function to load data into customeraggregatespend table

Error Handling and Logging: Same as TR-DLT-001

TR-DLT-006: Implement Delta Live Tables

Related Functional Requirement(s): FR-DLT-006

Objective: Implement Delta Live Tables to load and process customer and order data, and create and maintain ordersummary and customeraggregatespend tables.

Target Cluster Configuration: Same as TR-DLT-001

Job Flow / Pipeline Stages:

- Implement Delta Live Tables pipeline to load and process customer and order data
- Use Delta Live Tables to create and maintain ordersummary and customeraggregatespend tables

Data Transformations / Business Logic: Same as previous TRs

Error Handling and Logging: Same as TR-DLT-001